WHAT IF?! EP 04

FREE CALCULATOR
you commented DEN

The Hibernation Calculator



How much fat would you need to sleep through winter? Plug in the days, get the kilos. Back-of-napkin maths, built on numbers from the reel.

THE FORMULA

2,000 kcal you burn a day (rough human average) **x** **150** days of winter (five months) **÷** **9,000** kcal in 1 kg of body fat (9 kcal per g) **=** **33 kg** of fat to bank

MATHS This is the no-tricks case: you lie there and burn the usual 2,000 kcal a day. A real hibernator burns far less, which is the whole trick.

WORKED TABLE: BY BODY WEIGHT (150-day winter)

Your weight	Fat needed	As % of you	With "75% less" torpor	% of you
50 kg	33.3 kg	67%	8.3 kg	17%
60 kg	33.3 kg	56%	8.3 kg	14%
70 kg	33.3 kg	48%	8.3 kg	12%
80 kg	**33.3 kg**	**42%**	**8.3 kg**	**10%**
90 kg	33.3 kg	37%	8.3 kg	9%
100 kg	33.3 kg	33%	8.3 kg	8%

The fat bill does not depend on your weight, only on how many days and calories. For scale, a typical bear loses about **15 to 30%** of its weight over winter, and mothers up to about 40%. The 75% column applies ESA's "up to 75% less food and water" astronaut-torpor estimate to this maths. That is my arithmetic, not a measured result.

BY WINTER LENGTH (2,000 kcal a day)

Winter	60 days	90 days	120 days	150 days	180 days
Fat needed	13.3 kg	20 kg	26.7 kg	33.3 kg	40 kg
With 75% less	3.3 kg	5 kg	6.7 kg	8.3 kg	10 kg



How a bear banks its fat

Before denning, bears eat roughly 15,000 to 20,000 kcal a day. At about 300 kcal a burger, that is around 60 cheeseburgers. Every day. That is hyperphagia, and it is the bear's version of the table above.



Sources Calorie maths: my arithmetic (2,000 kcal x 150 days / 9 kcal per g) · ESA Exploration Science (explorationscience.esa.int/?p=471) · NPS Katmai bear hibernation blog · Denali.org, Mass Audubon (bear weight loss)

Bear vs you

Why a bear can do it and you would need a miracle. Plus your own calculator.



	Bear	You (in bed)
Heart	from about 55 beats a minute to as few as 9 (winter average about 14)	no big slowdown, sadly
Breathing	around 1 to 2 breaths a minute in deep hibernation	normal
Food	eats up to 20,000 kcal a day before denning (about 60 cheeseburgers' worth)	2,000 kcal a day (rough average)
Muscle	recycles urea into protein, so muscle barely shrinks	muscle and bone waste with long bed rest
Blood clots	low HSP47 in platelets lowers clumping (Science, 2023)	clot risk rises with months in bed
Toilet	mostly none: a plug of dried poop and licked-off hair	bad news
Length	a few weeks (Mexico) to 6+ months (Alaska)	0 safe days

Bears dodge clots, but they are "light" hibernators, easily woken, with body temperature down only a few degrees. Not every bear skips the toilet: some do go by month 6 or 7. The human column is plain physiology, not a study.

THE HUMAN WHO ALMOST DID

Japan, 2006. A hiker stranded on Mt. Rokko was found after about 24 days with a body temperature near 22 °C, almost no pulse and multiple organ failure. He recovered after about 50 days in hospital. Doctors called it torpor-like; a later review said it cannot be called hibernation.



THE SPACE PLAN

ESA says torpor could cut an astronaut's food and water needs by up to 75%, with the crew napping for most of a ~180-day trip to Mars. First human trials may be possible by the mid-2030s (one ESA researcher's estimate). Torpor in people is still experimental, and no safe method exists.



YOUR TURN

kcal a day _____ x days _____ ÷ 9,000 = _____ kg of fat

then x 0.25 for the "75% less" dream: _____ kg

SEE A DOCTOR IF you are sleeping far more than usual and still exhausted, you feel unusually cold, drowsy or confused, or you are losing weight without trying. Do not try to fast or sleep through winter: human torpor is experimental research, not a lifestyle.

Sources Tøien et al., Science 2011 (ScienceDaily; PBS NewsHour) · The Scientist, 2023 (HSP47 clots) · bear.org (fecal plug) · Mail & Guardian, Dec 2006 (Mt. Rokko hiker) · ESA Exploration Science; Space.com · NPS Katmai